# Communication-Aware Cooperative Agents

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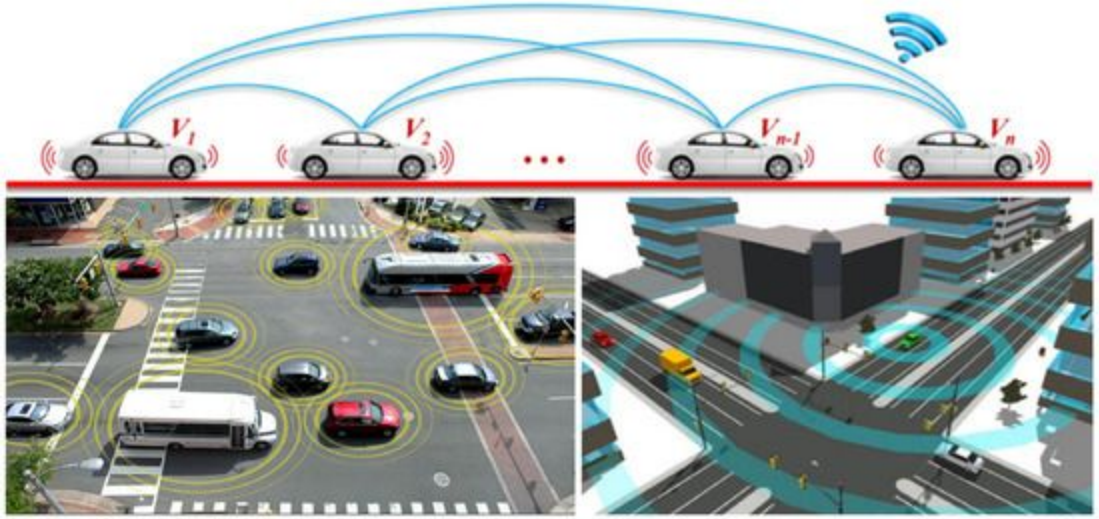
# Introduction

- V2X communications allow agents to share dynamic observations, trajectories, and localized intentions.
- However, physical wireless networks impose constrained bandwidth, operational delays, and channel noise.
- Evaluating how MARL maintains safety under imperfect communication remains a critical domain challenge.



# Analysis Dimensions

### What & When?

Do agents continuously broadcast state/action vectors, or use event-triggered mechanisms to send compressed messages?

### Channel Realism

Is communication assumed ideal, or are non-stationary network perturbations (latency, packet drop) integrated?

### Cost & Scalability

Is message bandwidth explicitly penalized, and how does the policy perform as vehicle density scales up?

# Literature Review

| Reference | Comm. Strategy Studied | Method | Key Finding | Limitations |
| --- | --- | --- | --- | --- |
| **Robust & Efficient Comm. in MARL** (survey, Liu et al. 2025) | Reviews everything: message corruption, delays, bandwidth limits, across many AI-team applications | Literature survey | The field almost always assumes perfect communication. | Admits no one tests **multiple** communication failures together at once — always one problem type at a time. Also admits driving-specific coordination under realistic network conditions is still unsolved. |
| **AgentComm-Bench** (Bansal & Gangwani, 2026) | Built a testing tool that breaks communication 6 different ways (delay, packet loss, bandwidth limits, async updates, stale data, conflicting data) | Toy grid-world tasks (not real traffic), tests 5 communication strategies including their own new method | Bad communication can actually hurt performance more than no communication at all in some cases; sending messages twice helps under packet loss | Purely toy grid-world games, not a real traffic simulator (SUMO/CARLA). Does not test failures combined together. Not built for transportation specifically. |
| **TMC** (Zhang et al., NeurIPS 2020) | Agents send a message only if it's meaningfully different from the last one (adaptive threshold) | Deep learning-based multi-agent team + message buffering | Cuts communication by up to 80% and still performs well even under heavy message loss | Tested only on a video game (StarCraft) and two toy grid tasks. Never tested with delay (latency), only message loss. Not a transportation task. |

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|---|---|---|---|---|
| **IntNet** (Parada et al., IEEE RA-L, 2025) | Vehicles sharing their *intent* (predicted future actions) with each other, plus a smart scheduler deciding who needs to hear from whom | Deep learning: Graph Attention Network processes incoming messages; scheduler dynamically prunes the communication graph; tested with mixed autonomous + human-driven vehicles in Highway, 4-way Intersection, and Lane Merging scenarios | Cuts communication load by ~60% while improving coordination and lowering collision rates compared to prior communicative MARL methods | No communication-failure testing at all — no packet loss, no latency, no bandwidth stress test. Assumes messages that do get sent always arrive perfectly. |
| **ETCNet** (Hu et al., IEEE TNNLS, 2023) | Deciding whether an agent should send or receive a message at each step, given a strict bandwidth budget | Reinforcement learning: bandwidth limit is turned into a "penalty" in the agent's objective; two versions built — one that learns when to *send* (ETSNet), one that learns when to *receive* (ETRNet) | Agents learn to communicate only when it's actually worth it, cutting bandwidth use while keeping team performance close to full-communication levels | Only tests bandwidth limits — no packet loss, no delay/latency, no message corruption. |

# Inferences

**Nobody has taken a real transportation coordination task, and systematically tested it against several communication failures combined together (loss + delay + bandwidth limits, etc  at once, not one at a time), using a simple, interpretable agent design. Every paper either:**

- **tests general AI teams, not transportation (TMC, AgentComm-Bench),**
- **tests only toy games, not real traffic networks,**
- **tests failures one at a time, not combined, or**
- **explicitly admits this combination hasn't been done (the survey).**

# Potential Research Questions

**RQ1 (primary, most defensible):**
*"In a simple rule-based cooperative transportation-coordination task, how does coordination performance and collision/conflict risk degrade under combined communication impairments (packet loss, latency, and bandwidth limitation, etc (potential) applied jointly) compared to each impairment tested in isolation?"*

- Hypothesis: Combined impairments degrade performance super-additively (worse than the sum of individual effects) because agents' error-correction/recovery strategies for one failure type are undermined by simultaneous exposure to another.
- Why it's solid: directly targets the survey's own stated gap ("multiple disruptions co-occur" is unstudied) and AgentComm-Bench's own admitted limitation (only tests impairments independently, never combined, and never on a real traffic network).

# Future Work

**Move towards RL frameworks, Deep RL frameworks, and incorporate human behaviour to correctly access the situations and take better decisions**

**Idea 1 — Predictive Event-Triggered Communication**

- **Agents run local prediction model (dead-reckoning/Kalman filter) of neighbors' trajectories**
- **Trigger comm only when actual state deviates > threshold (ε) from predicted state**
- **On packet loss: use prediction until safety deadline → fallback to conservative braking**

**Idea 2 — Criticality-Aware Reliable Retransmission**

- **High-severity events (e.g., emergency braking) require explicit ACK/heartbeat from neighbors**
- **If ACK dropped: retransmit via exponential backoff / dynamic power boost**
- **Trade-off: small comm-cost spike, but prevents "single dropped packet → crash" failure**

# Thank You

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